

ERROR SURFACE & FAILURE MODE ANALYSIS

Arabic Fake News Detection

Student: Abdullah Abbas Ezzat | ID: 2405415 | Supervised by Prof. Mohamed M.Abbassy

— MODULE 01 —

THE APPROACH

Don't celebrate the 99% – autopsy the 1%

\$ Why Failure Mode Analysis?

— The Mindset Shift

▶ We don't look at the 91.28% success rate – we autopsy the 8.72% failure rate

— What we're looking for:

▶ Is the model failing on content – or on structural artifacts?

▶ Does it predict 'Fake' just because it sees the word 'Urgent'?

▶ # # That's a Spurious Correlation, not real understanding

▶ → Map the error surface → build guardrails for production

— MODULE 02 —

KEY METRICS

What we measure and why

\$ The 3 Diagnostic Metrics

METRIC

- ▶ Error Density (ρ)
- ▶ HCFR
- ▶ Feature Importance Shift

WHAT IT ANSWERS

- ▶ Where do errors cluster by feature range?
- ▶ How often is the model wrong AND confident?
- ▶ Do error words differ from correct-prediction words?

— MODULE 03 —

MODEL OVERVIEW

MARBERTv2 – strong numbers, hidden problems

\$ MARBERTv2 – Quick Results

Overall Accuracy → ~99.1%

True Positives → 35,365 (Fake → Fake)

True Negatives → 11,609 (Real → Real)

False Negatives → 302 (Fake slipped through as Real)

False Positives → 209 (Real flagged as Fake)

ROC AUC → 0.565 ← this is the problem

99% accuracy + AUC of 0.565 = overconfident and poorly calibrated

— GRAPH 01 —

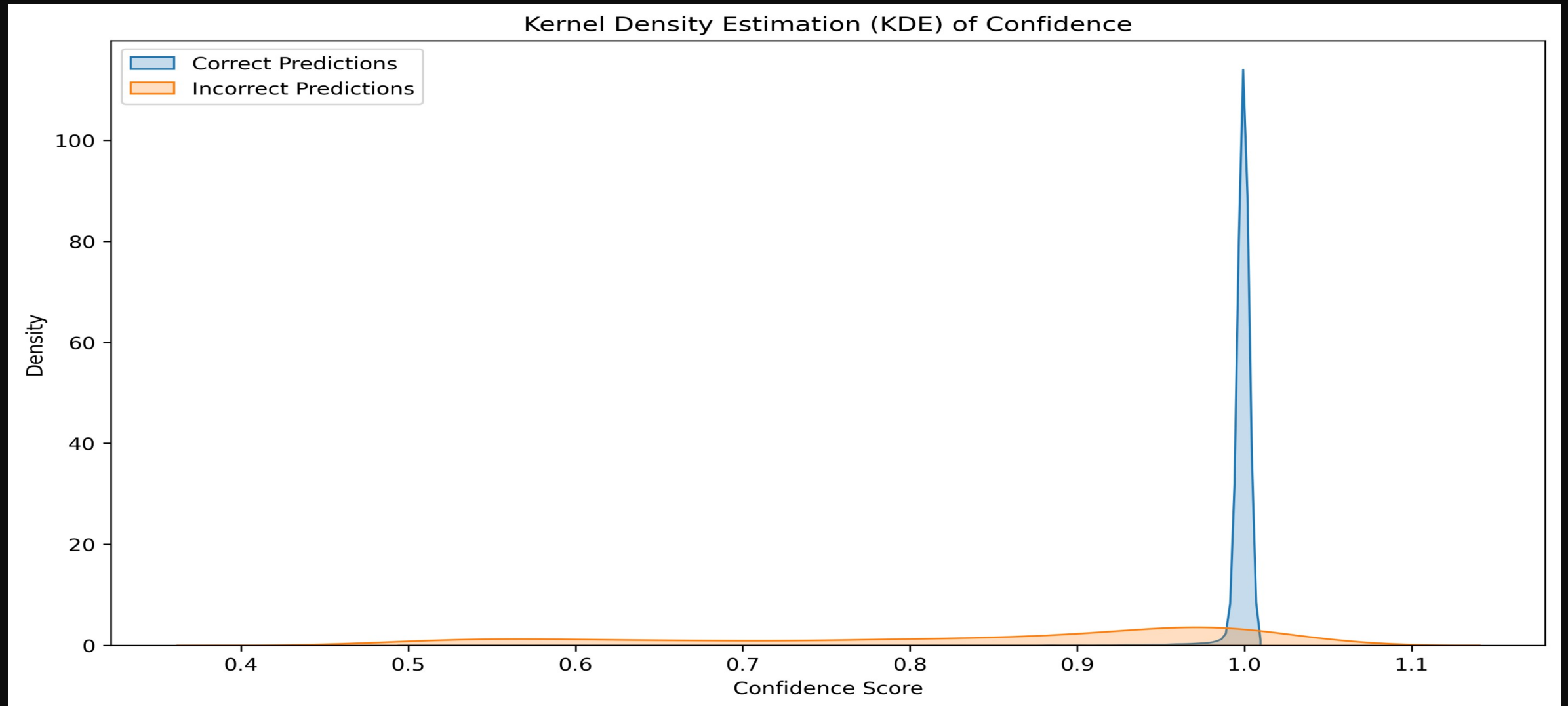
CONFIDENCE KDE

Correct vs Incorrect — how confident is the model?

\$ How to Read the KDE Plot

- ▶ KDE = Kernel Density Estimation – shows the confidence score distribution
 - ▶ Blue peak (correct) → sharp spike at 1.0 → model is near-certain when right
 - ▶ Orange curve (incorrect) → flat, spreads from ~0.4 all the way to 1.0
- ▶ The flat orange tail near 1.0 is the real problem
- ▶ Model is making wrong calls while still being highly confident
- ▶ # # High confidence $\neq$ correct prediction → calibration issue before deployment

Confidence KDE – Correct vs Incorrect Predictions



Blue spike at 1.0 = correct predictions | Orange flat tail = wrong but confident

— GRAPH 02 —

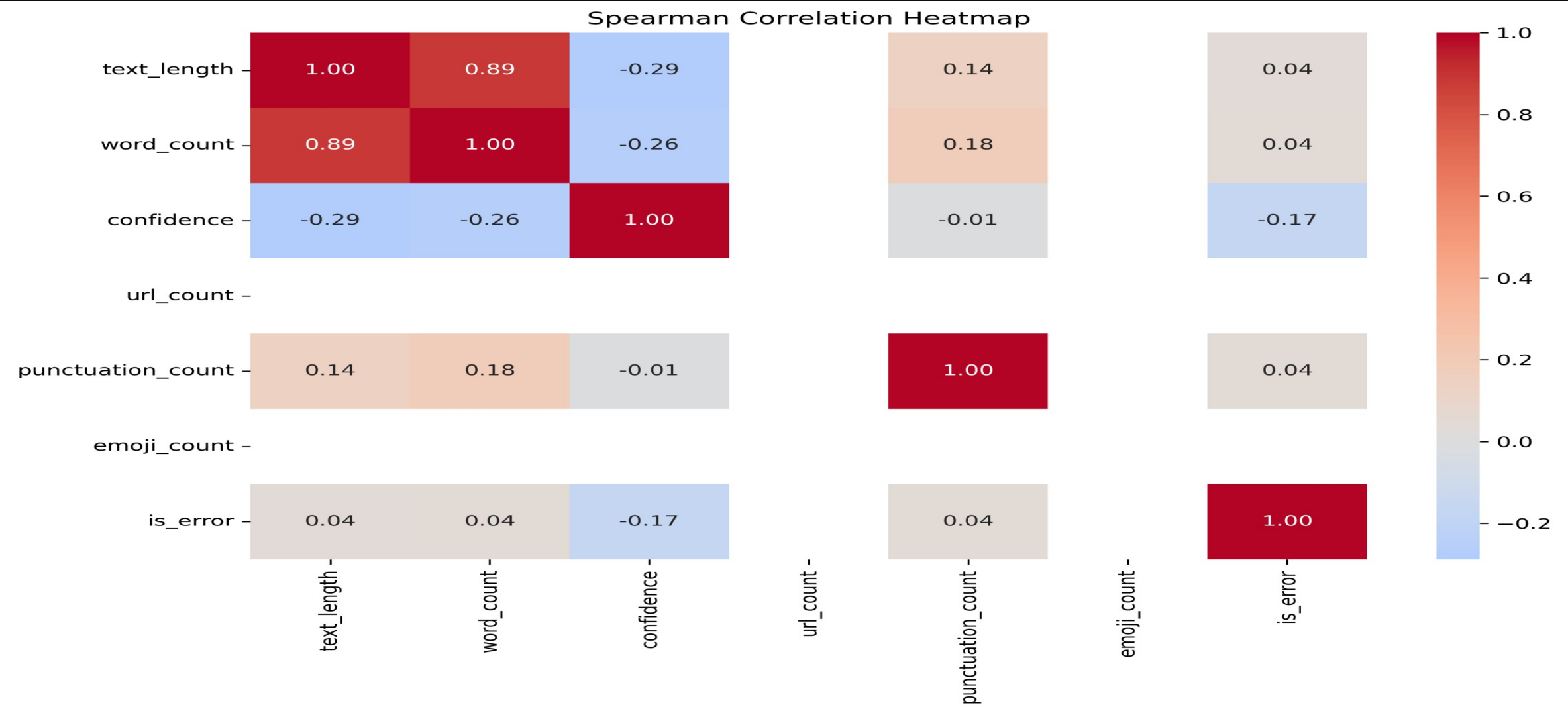
SPEARMAN CORRELATION HEATMAP

Which features actually matter?

\$ What the Heatmap Shows

- ▶ `text_length` vs `word_count` = 0.89 → nearly the same feature, redundant
- ▶ Confidence has mild negative correlation with length (-0.29, -0.26)
- ▶ → Model gets slightly less confident on longer texts – makes sense
- ▶ `is_error` vs confidence = -0.17 → weak but there, errors happen at lower confidence
- ▶ `URL count` and `emoji count` → near zero correlation with everything
- ▶ # # `URL` and `emoji` features carry almost no signal for this task

Spearman Correlation Heatmap



text_length ↔ word_count = 0.89 (redundant) | URL & emoji = no meaningful signal

— GRAPH 03 —

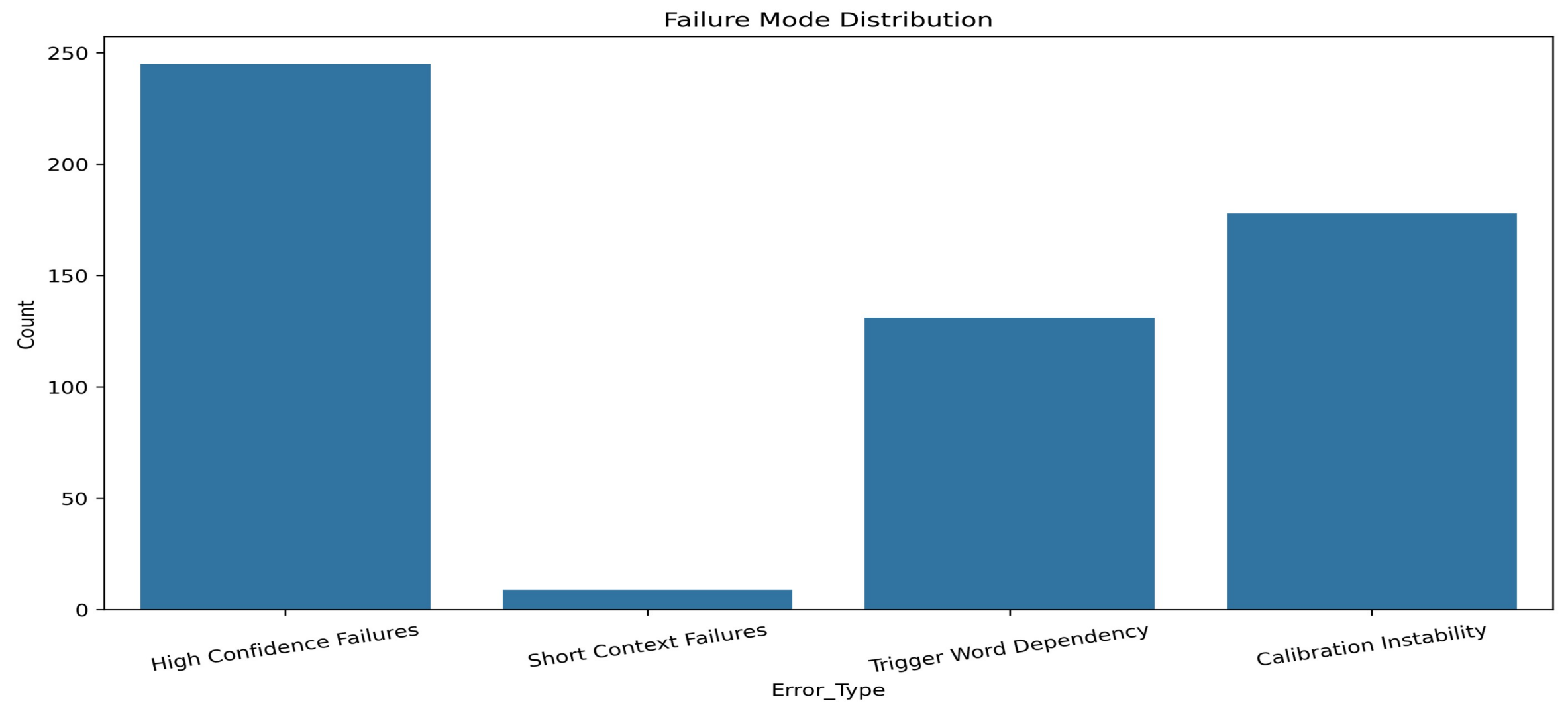
FAILURE MODE DISTRIBUTION

What kind of errors does the model make?

\$ Breaking Down the 4 Error Types

- ▶ High Confidence Failures → ~246 cases ← most dangerous
- ▶ Model is wrong and doesn't know it – hardest to catch without post-processing
- ▶ Calibration Instability → ~178 cases
- ▶ Inconsistent confidence scores for similar inputs
- ▶ Trigger Word Dependency → ~131 cases
- ▶ Model leans on specific words instead of understanding context
- ▶ Short Context Failures → ~9 cases (barely relevant)
- ▶ # # Takeaway: most failures are confidence-related, not comprehension-related

Failure Mode Distribution



High Confidence Failures dominate (~246) | Short Context almost irrelevant (~9)

— GRAPH 04 —

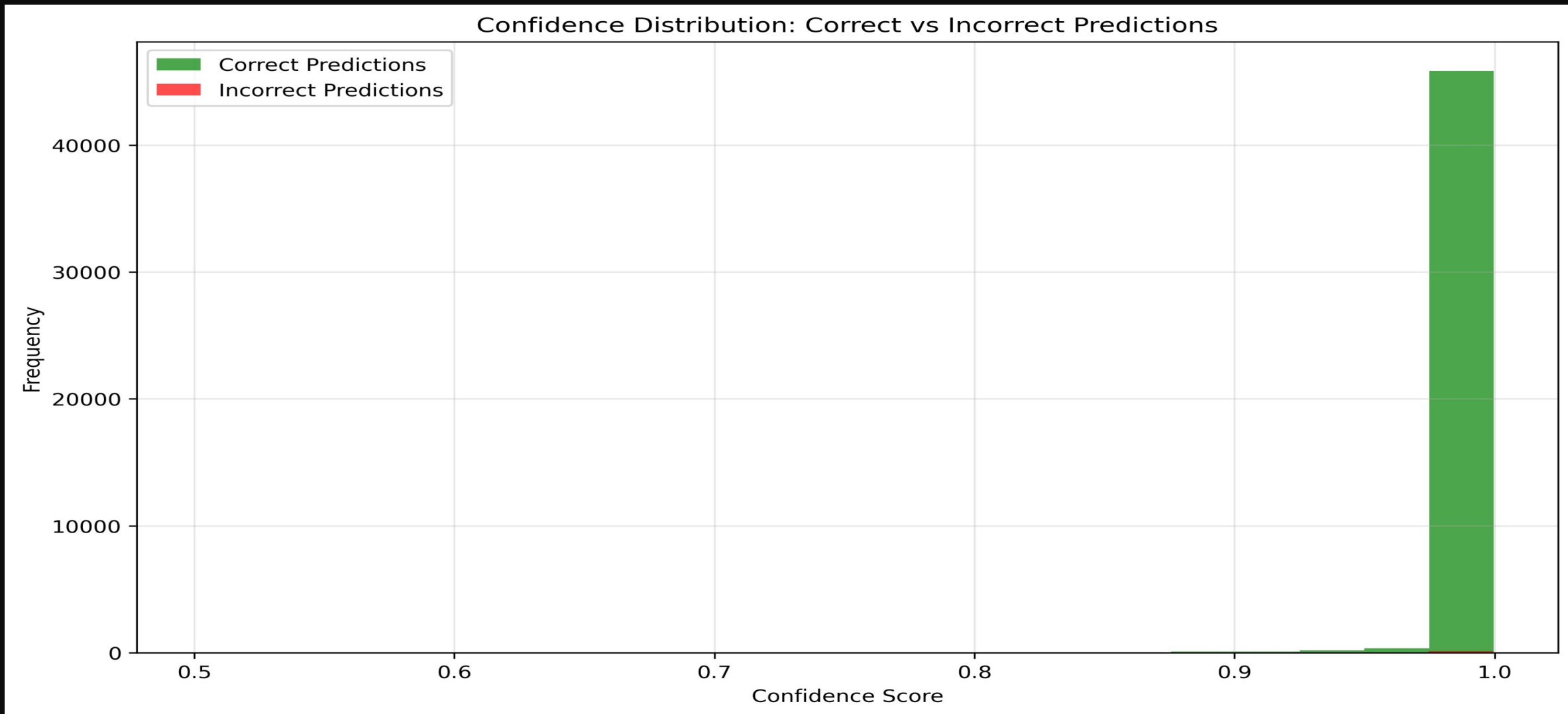
CONFIDENCE DISTRIBUTION

MARBERTv2 – what does the histogram tell us?

\$ MARBERTv2 Confidence Distribution

- ▶ Almost everything is stacked in the 0.95–1.0 bin
- ▶ Correct predictions (green) → over 45,000 cases in that last bin
- ▶ Incorrect predictions (red) are barely visible – swamped by correct volume
 - ▶ This is why raw accuracy looks great – it's hiding class imbalance
- ▶ The model predicts one class so frequently that the numbers lie to you
 - ▶ # # High accuracy here = misleading – always check confidence distribution too

MARBERTv2 Confidence Distribution – Correct vs Incorrect



45,000+ correct in 0.95-1.0 bin | Errors invisible – buried under volume

— GRAPH 05 —

CONFUSION MATRIX

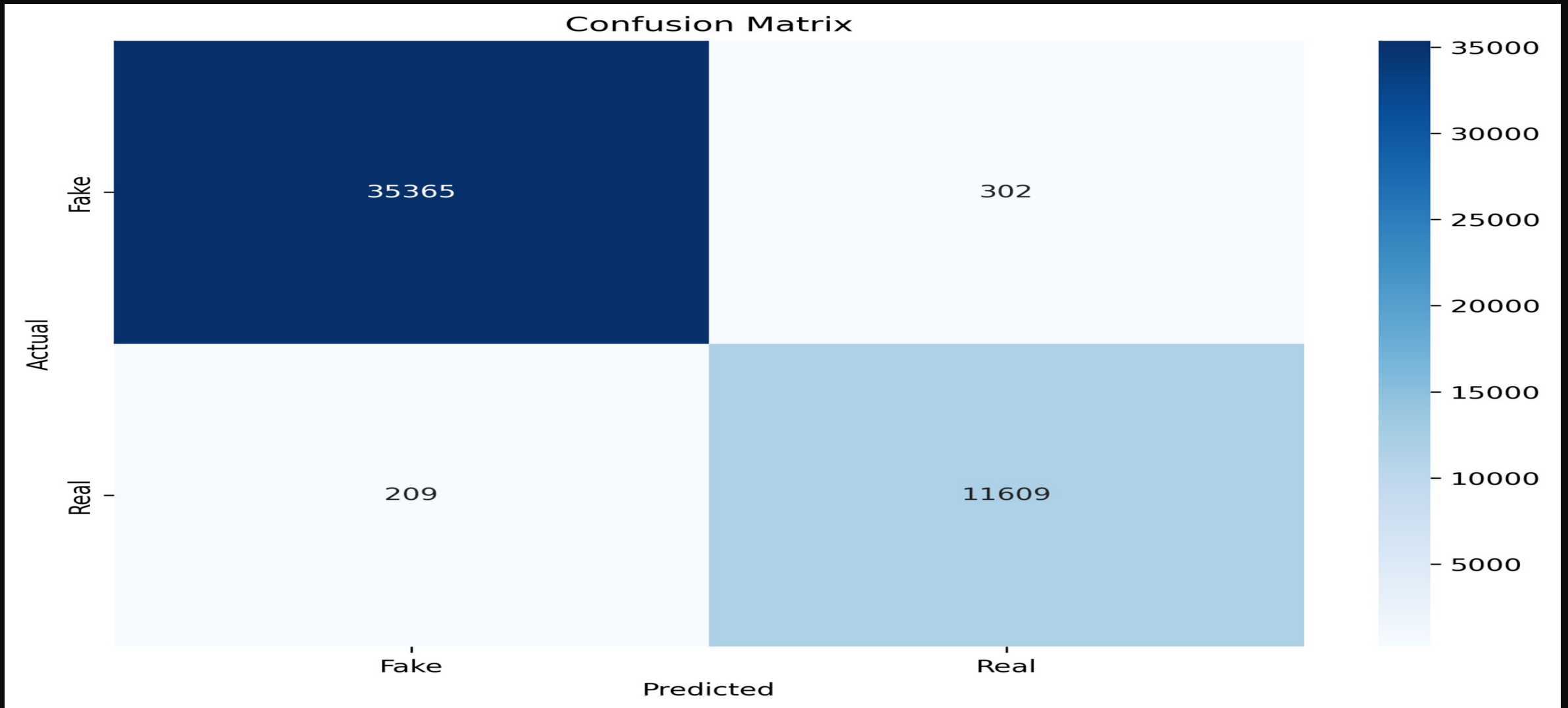
Where exactly does the model go wrong?

\$ Confusion Matrix – The Numbers

True Positives	→	35,365	Fake correctly caught
True Negatives	→	11,609	Real correctly passed
False Negatives	→	302	Fake slipped through as Real ← costly
False Positives	→	209	Real flagged as Fake

In misinformation detection: False Negatives are the more dangerous mistake
→ 302 fake articles reached readers labeled as real news

Confusion Matrix – MARBERTv2



TP=35,365 | TN=11,609 | FN=302 (fake missed) | FP=209 (real flagged)

— GRAPH 06 —

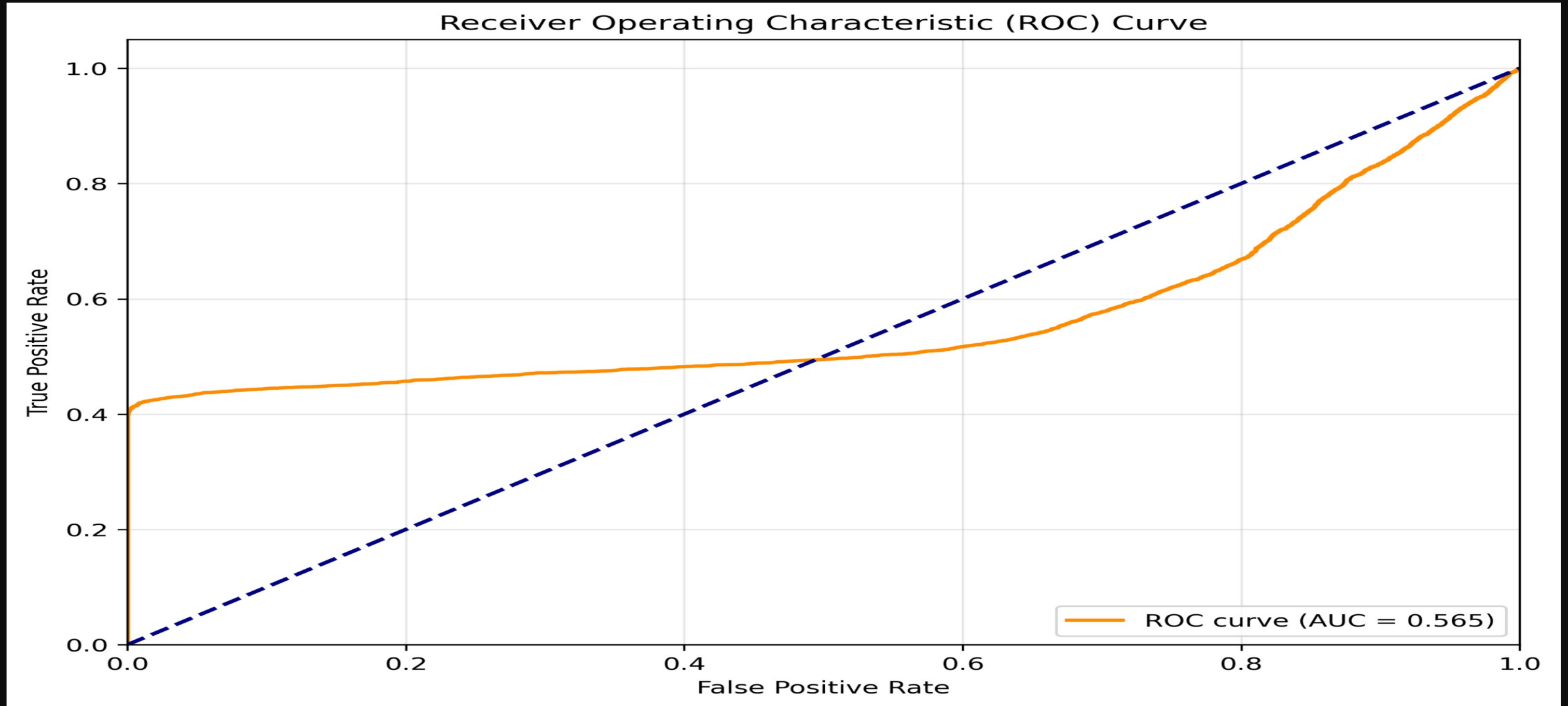
ROC CURVE — AUC = 0.565

The metric that exposes the real problem

\$ Why AUC = 0.565 is a Red Flag

- ▶ A 99% accuracy model should NOT have an AUC of 0.565
- ▶ Random classifier = 0.5 | Perfect classifier = 1.0
- ▶ 0.565 = barely above random → confidence score is not a reliable signal
- ▶ ROC curve hugs the diagonal for most of its range – only pulls away top-right
 - ▶ You cannot use confidence as a threshold to flag uncertain cases – it won't work
 - ▶ # # Direct consequence of the calibration failures seen in Graph 1 and Graph 3

ROC Curve – AUC = 0.565



Curve hugs the diagonal → confidence score barely separates errors from correct predictions

— GRAPH 07 —

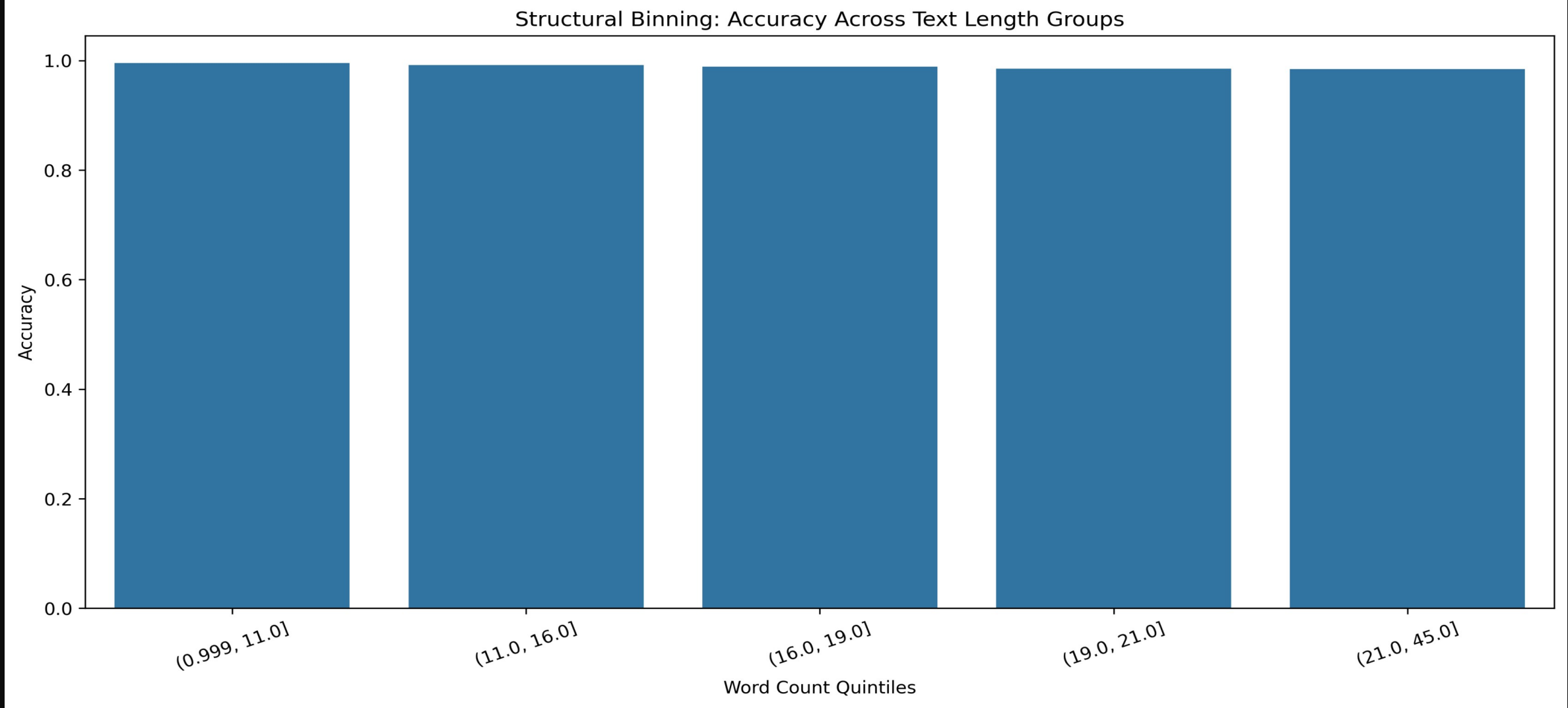
STRUCTURAL BINNING

Does text length cause errors?

\$ Accuracy Across Text Length Groups

- ▶ Dataset split into 5 word-count quintiles: from <11 words up to 45 words
- ▶ Accuracy stable across all 5 groups: between 99.3% and 99.8%
- ▶ Slight downward trend as texts get longer – but it's marginal
- ▶ No length range where the model clearly struggles
- ▶ → Text length is NOT a meaningful source of errors here
- ▶ # # This rules out short-context weakness as a driver of failures

Structural Binning: Accuracy Across Text Length Groups



All 5 quintiles sit between 99.3–99.8% | Text length does not drive errors

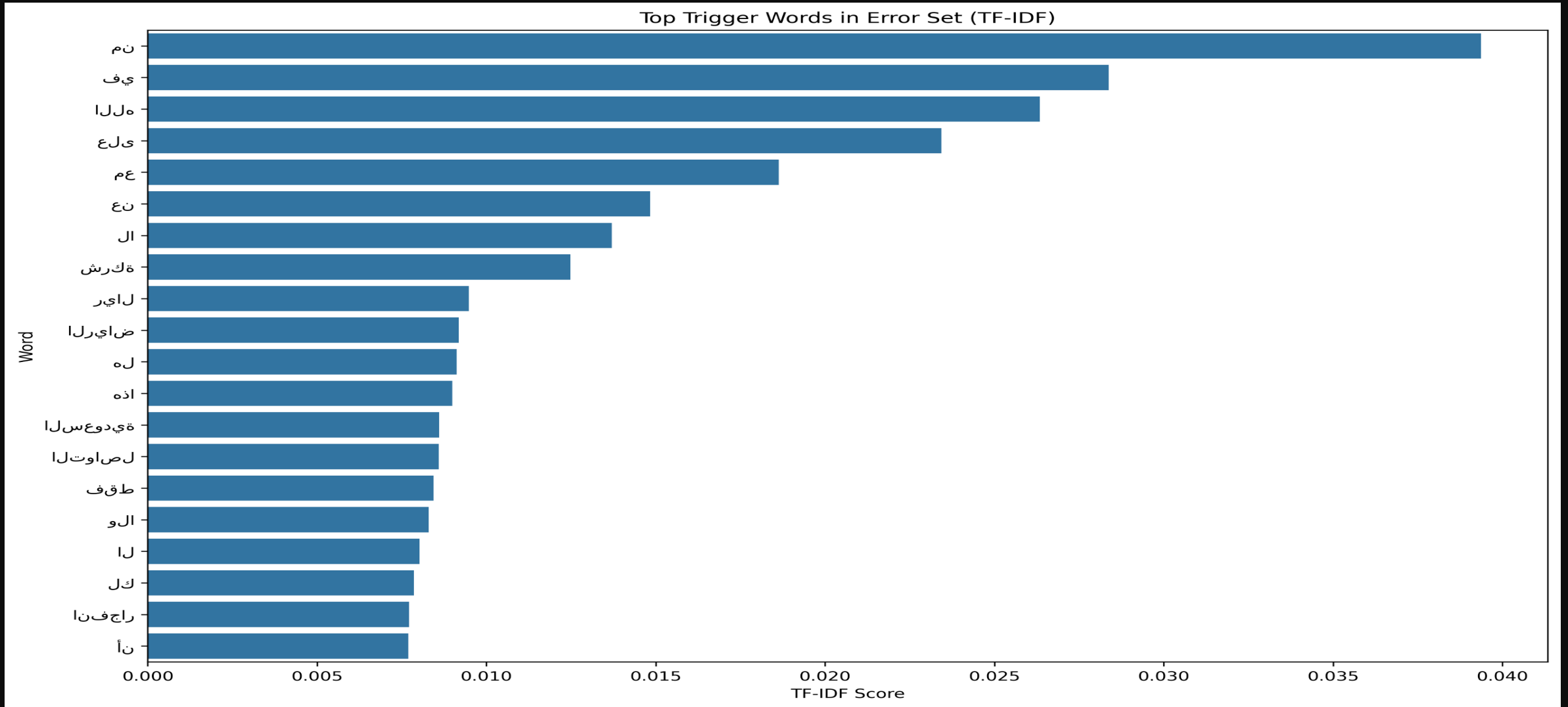
TRIGGER WORDS (TF-IDF)

What words appear most in the error set?

\$ Top Trigger Words – What They Tell Us

- ▶ Top words are Arabic prepositions and function words – normally TF-IDF downweights these
 - ▶ High scores for stop words → error set may lack stop-word filtering
 - ▶ Lower on the list: sharika (company), riyal, Riyadh, Saudi Arabia
 - ▶ → Cluster of errors around Saudi financial and business content
 - ▶ Model likely hasn't seen enough of that domain during training
- ▶ ## Fix: targeted data augmentation with Saudi financial/business Arabic news

Top Trigger Words in Error Set (TF-IDF)



Top = Arabic prepositions | Lower = Saudi financial terms (company, riyal, Riyadh)

— CONCLUSION —

FINAL VERDICT

What does all of this actually mean?

\$ Key Takeaways

- 99% accuracy is real – but misleading
 - ▶ AUC of 0.565 exposes that confidence scores are barely above random
- Most failures are confidence-related, not comprehension-related
 - ▶ 246 high-confidence failures – wrong AND certain – hardest to catch
- There is a domain gap around Saudi financial Arabic content
 - ▶ Trigger word analysis points to a fixable problem with targeted augmentation
 - ▶ # # Production fix: flag predictions with confidence in the 0.7–0.95 range for human review

[EOF]

Abdullah Abbas Ezzat | ID: 2405415 | Questions welcome

ezz@terminal:~# _